EFY TESTED
S.C. Dwivedi

GPS Based **Dot-Matrix Display Clock**

Kamal Das

Presented here is an LED based 2D matrix display clock. There are a number of technologies used to create various displays in use today. The basic display systems we know are segment displays, 2D displays, 3D displays, and also mechanical displays. The LED display is a flat panel display which uses an array of LEDs as pixels. This project is for making a GPS based LED dot-matrix clock. The EFY Lab's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the GPS based dot-matrix display clock is shown in Fig. 2. It uses two Arduino Nano boards (Board1 and Board2), a NEO 6M GPS module, and a readymade 8x8 dot-matrix display board.

Note that the system works with two different Arduino boards, one of which (Board2) controls the dot-matrix display and the other (Board1) generates serial clock data from GPS module at TX1 pin and feeds to RX0 pin of Board2.

The project has four sections: 8x8 dot-matrix display, display driver, interfacing of Arduino Nano board with dot-matrix, and GPS module.

Dot matrix display. The readymade 4-in-1, 8x8 display board (DIS1) shown in Fig. 3 is well integrated and comes with five pins (Vcc, GND, CLK, CS, and Data). The whole module comes in four 8x8 common-cathode dot-matrix form with each dot-matrix driven by MAX7219 IC.

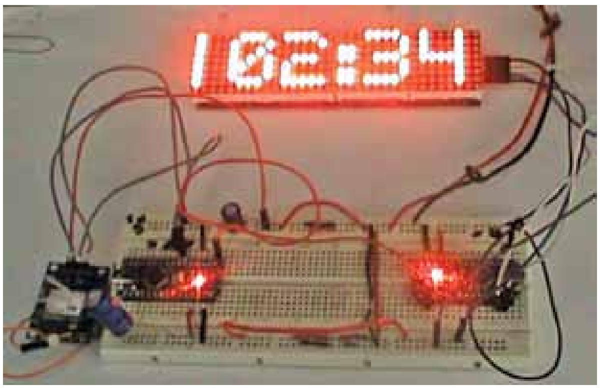

Fig. 1: EFY lab's prototype

Display driver. The display driver IC MAX7219 is from Maxim Integrated whose operating voltage is 4V to maximum of 5.5V. Its pin 19 is for power supply, pins 4 and 9 are for ground, pin 1 for Data In, pin 12 for Load, and pin 13 for Serial Clock. (The pin details are not shown here.) Pin 24 is Data Out pin for forwarding the data to other segments.

Interfacing Arduino with display. Arduino Nano (Board2) is preferred for interfacing; it has a USB interface to connect to PC or laptop. First install Arduino IDE and connect Arduino Nano to USB port of the PC. Open the Arduino sketch/program (EFYlabmatrixdisplay.ino), then include libraries from Manage Libraries option. Search for *MD_Parola 2.7.4* and *MD_MAX72XX 2.10.0* libraries and install them. Now, upload the Arduino code (*EFYlabmatrixdisplay.ino*) to Board1. As you can see from circuit in Fig. 2, CLK pin of display module (DIS1) is connected to pin 13 (D13) of Board2, Data pin to pin 11, CS pin to pin 10, GND to GND pin, and Vcc pin to +5V pin of Arduino Nano (Board2).

Interfacing Arduino with GPS. Here the other Arduino Nano (Board1) is used for calculating and parsing the GPS data received through the NEO 6M GPS module. The program (*efylabGPSclock.ino*) written in Arduino programming language is compiled using Arduino IDE. Open Arduino IDE, open the sketch, then include library from Manage Libraries option. Search for tinyGPS++ by Mikal Hart that provides object oriented analysing of NMEA data to Arduino. Click install Tiny GPS++ from Manage Libraries window. You also need softwareSerial

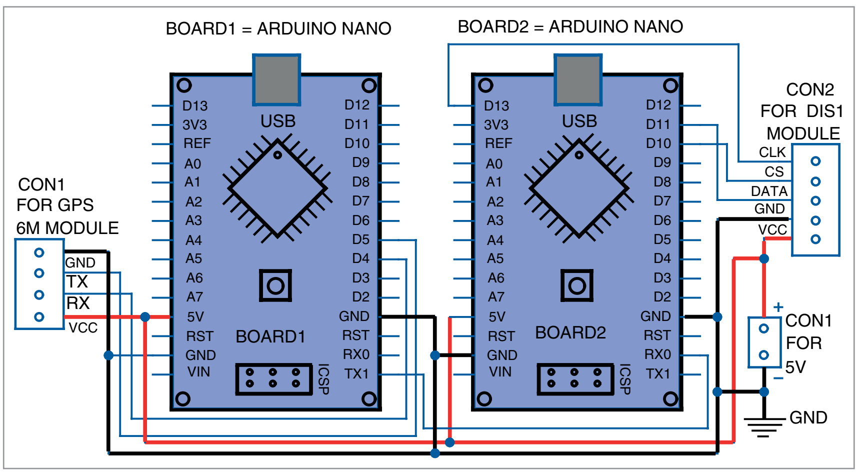

Fig. 2: Circuit diagram of GPS based dot-matrix display clock

EFY Note
The source code of this project is available for free download at *source.efymag.com*